



DR. HARSHIL SHAH
ORTHOPAEDIC SURGEON

KNEE SPORTS INJURY

**ACL, MENISCUS &
ARTHROSCOPIC CARE**

Evidence-based. Patient-focused.
Performance-driven recovery.



EXPERT
CARE



ADVANCED
TECHNOLOGY



PERSONALISED
TREATMENT



FASTER, SAFER
RECOVERY

CONSULTATION ARTHROSCOPY RECONSTRUCTION REHABILITATION

01/06

ARTHROSCOPY & ACL INJURY

What happens, why surgery may be needed.



WHAT IS ARTHROSCOPY?

A minimally invasive technique using a small camera and specialised instruments to examine and treat problems inside the knee joint such as cartilage injuries, meniscus tears, loose bodies or ACL injuries.

COMMON ACL TEAR CAUSES

- ✓ Pivoting or twisting on planted foot
- ✓ Sudden deceleration or change in direction
- ✓ Landing awkwardly after a jump
- ✓ Direct impact to the knee
- ✓ Overstretching or hyperextension
- ✓ Degeneration over time

DOES EVERY ACL TEAR NEED SURGERY?

No. Some low demand patients with a stable knee and minimal symptoms may be managed with structured physiotherapy.

Reconstruction is more commonly recommended for active patients, athletes, or those with instability or recurrent giving way episodes.



THERE IS NO SINGLE "BEST" GRAFT

Choice depends on age, sport, occupation, activity level and surgeon experience. The aim is to match the graft to the patient and their return-to-activity goals.

ACL RECONSTRUCTION

Graft choices & extra stabilisation (LET)

GRAFT OPTIONS



**Hamstring
Autograft**

Lower donor site
pain/weakness



**Bone-Patellar
Tendon-Bone
Autograft**

Strong fixation,
bone-to-bone healing



**Quadriceps
Autograft**

Thick, strong graft,
increasingly
used



Allograft

Donor tissue
Prefer for older
or less active
patients



ACL + LATERAL EXTRA-ARTICULAR TENODESIS (LET)

- LET adds an extra stabilising "checkrein" on the outside of the knee to control rotational instability.
- Often considered in higher-risk patients – pivoting sports or those with persistent instability.

GRAFT CHOICE CONSIDERATIONS

Factor	What we consider
Age & Activity	Younger, high-demand athletes may benefit from autograft
Sport	Pivoting/contact sports → higher-rate risk if LET
Knee Stability	Associated injuries, limb and alignment
Previous Surgery	Revision ACL or multi-ligament surgery
Patient Goals	Return to sport level and timeline



OUR GOAL

Restore stability,
protect the knee
and get you back
to doing what
you love.

MENISCUS TEARS

Understand, heal or treat.



MENISCUS ZONES

Blood supply determines healing potential

- **RED – RED**
 Good blood supply.
 Best chance to heal.
- **RED – WHITE**
 Some blood supply.
 Healing possible.
- **WHITE – WHITE**
 Poor blood supply.
 Low healing potential.

TEAR PATTERNS

TYPICAL FEATURES

**Longitudinal /
Bucket Handle**

Can flip into the joint and
cause a locked knee

Radial

Cuts across meniscus fibers
and reduces shock absorption

Flap / Horizontal

Often more degenerative,
may create a unstable flap

Root Tear

Disrupts meniscus attachment
and can severely compromise
knee joint function

REPAIR VS TRIMMING

When tear pattern, tissue quality and blood supply allow, we repair the meniscus to preserve knee function and protect the joint long term.



TRIMMING (PARTIAL MENISCECTOMY)

When tear is not repairable or in degenerative tears, trimming the damaged part relieves symptoms.



OUR AIM: Preserve as much healthy meniscus as possible.

REHABILITATION & RETURN TO RUNNING

Progressive phases for safe recovery.

- 1 Phase 1 – Protection (0–2 weeks)**
 - Control swelling & pain
 - Activate quadriceps
 - Restore knee extension
 - Gentle range of motion
- 2 Phase 2 – Early Strength (2–6 weeks)**
 - Build stable gait & balance
 - Hip & core strength
 - Straight leg raise
 - Closed-chain exercises
- 3 Phase 3 – Progress Strength (6–12 weeks)**
 - Increase load & function
 - Introduce light impact activities
 - Strength & neuromuscular control
- 4 Phase 4 – Running Control (12–20 weeks)**
 - Agility drills
 - Plyometrics
 - Sport-specific patterns
- 5 Phase 5 – Return to Sport (6+ months)**
 - Sport-specific testing
 - Confident return with control
 - Performance testing



IMPORTANT MODIFICATIONS

Timelines vary. Progress is based on quality of movement, strength and control – not just time. Your surgeon and physiotherapist will guide each step.

PRACTICAL QUESTIONS

Common queries, milestones & return to sport.



When can I walk?

Usually within days, with crutches as advised by your surgeon.



When can I drive?

Typically 3–6 weeks, once you have good control and are off pain medication.



When can I run?

Straight-line running often begins around 12–16 weeks as per plan.



When can I play sport?

Straight-line running often begins around 3–6 months when criteria are met.

RETURN TO SPORT GUIDANCE



4–6 Months

Low-impact sports



7–9 Months

Agility & change of direction



9–12 Months+

Full sport participation

KEY PRINCIPLES

- ✓ Individual recovery, not just calendar time
- ✓ Strength symmetry $\geq 90\%$
- ✓ Good movement quality & confidence
- ✓ Clearance with functional tests by your team



OUR GOAL

Restore your knee, rebuild your confidence and help you return to the sport you love – safely and sustainably.



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Compassion. Precision.
Excellence.